

Visualizing Tile-based Vector Data on 3D Terrain

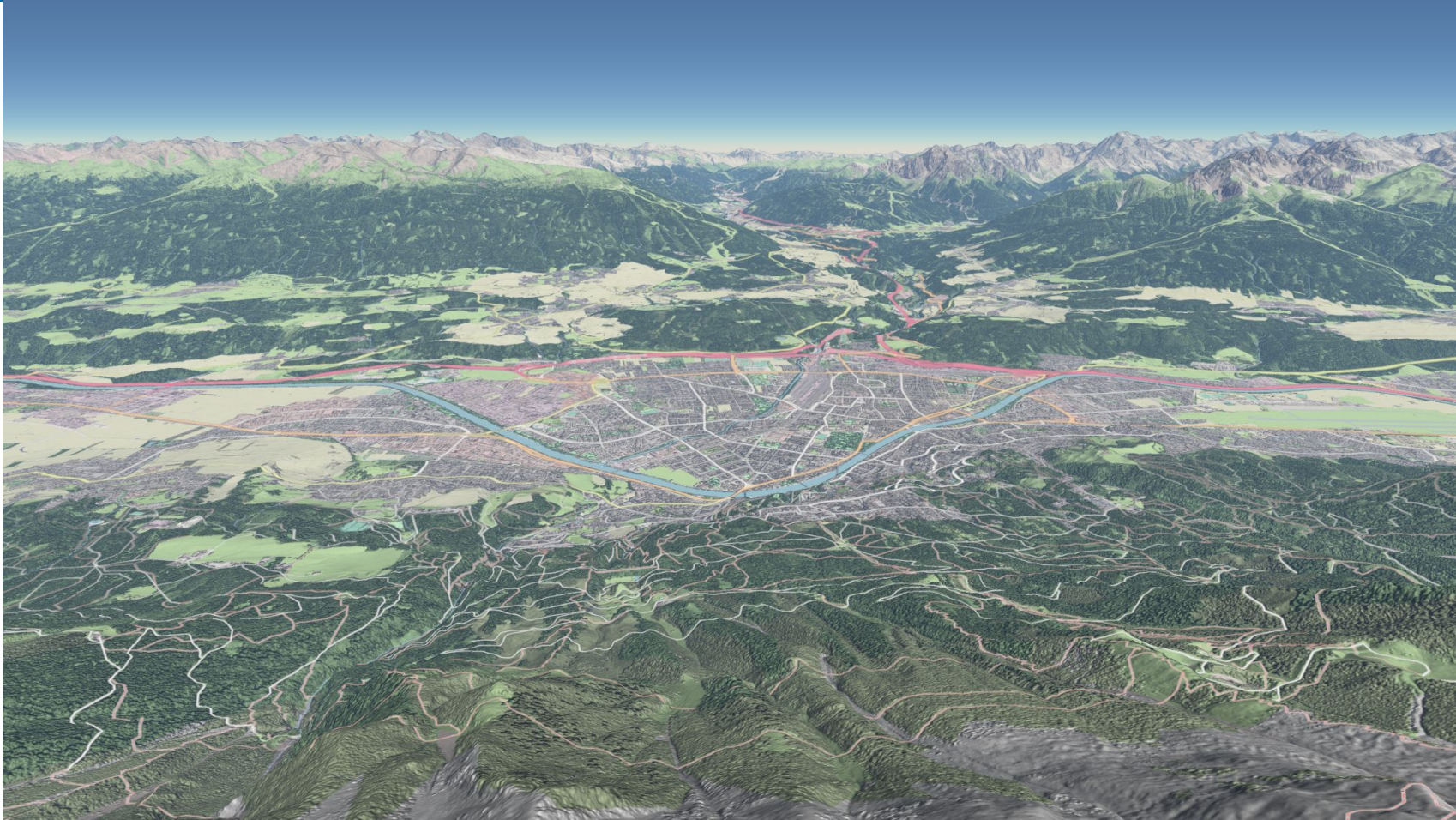
Lucas Dworschak

1225883

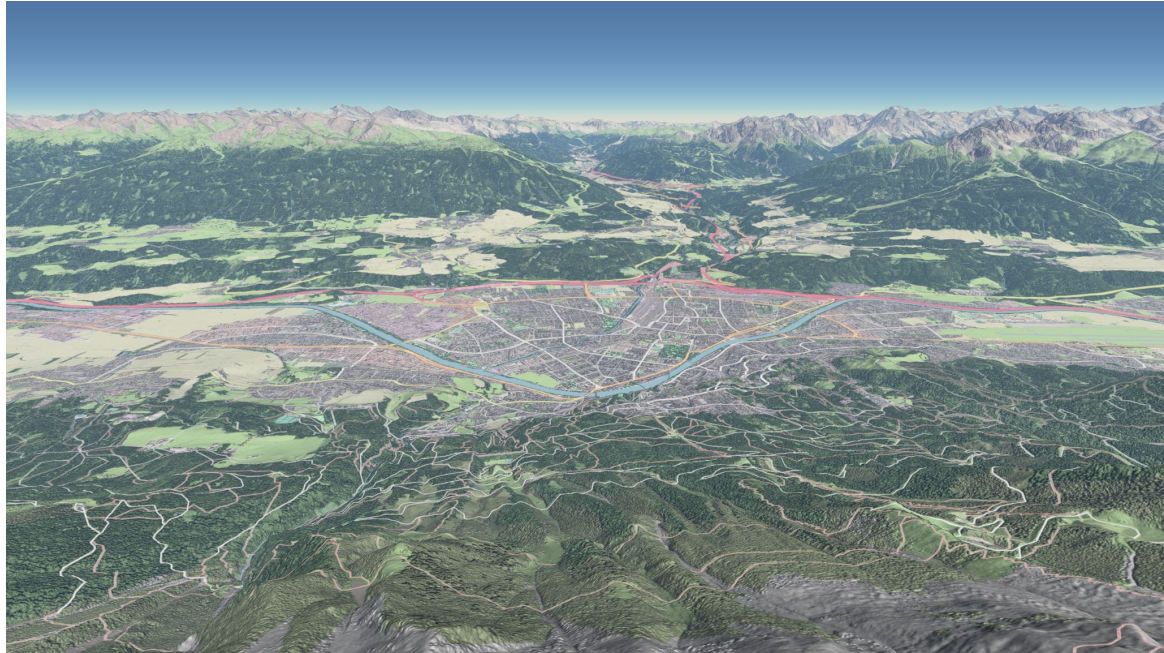
Research Unit of Computer Graphics
Institute of Visual Computing & Human-Centered Technology
TU Wien, Austria



Visualizing Tile-based Vector Data on 3D Terrain



- Visualizing Geospatial Data
 - Scalable
 - Precise
 - Performant
 - Storage efficient



- Visualizing Geospatial Data
 - **Scalable**
 - Precise
 - Performant
 - Storage efficient



Image source: <https://docs.maptiler.com/google-maps-coordinates-tile-bounds-projection/>



- Visualizing Geospatial Data
 - Scalable
 - **Precise**
 - Performant
 - Storage efficient



- Visualizing Geospatial Data
 - Scalable
 - Precise
 - **Performant**
 - Storage efficient

Scene	Schneeberg	Innsbruck	Vienna Side	Vienna Top
Geometries	2,263,901	3,396,163	4,482,712	7,082,781
Raster	2.99 ms	3.34 ms	3.02 ms	2.54 ms
Vector	12.03 ms	16.51 ms	20.04 ms	24.53 ms

Rendered with AMD Radeon RX 580 Series, on 1920 x 1080 resolution



- Visualizing Geospatial Data
 - Scalable
 - Precise
 - Performant
 - **Storage efficient**

Buffer	Bits / texel	W x H x N	MB
Style	64	128 x 128 x 1	~0.1
Elevation	16	65 x 65 x 2048	~16.5
Acceleration grid	32	64 x 64 x 512	8
Geometry (128k)	64	128 x 128 x 512	64
Geometry (256k)	64	256 x 256 x 512	256
Geometry (512k)	64	512 x 512 x 128	256
Fallback (high res.)	16	512 x 512 x 512	256
Fallback (low res.)	16	256 x 256 x 512	64
Surface shading	4	512 x 512 x 512	64
Total			~ 984



- Thöny et al.
 - Deferred vector map visualization [TBP16]
 - Large-scale pixel-precise deferred vector maps [TBP18]
- Amiraghdam et al.
 - Localis [ADP20]
 - Loops [ADP22]
- Zechmeister [Zec20]



- Used in 2D maps
- Vector vs Raster
- Hierarchical
- Zoom dependent Visualization
- Web Mercator Projection



Image source: <https://docs.maptiler.com/google-maps-coordinates-tile-bounds-projection/>

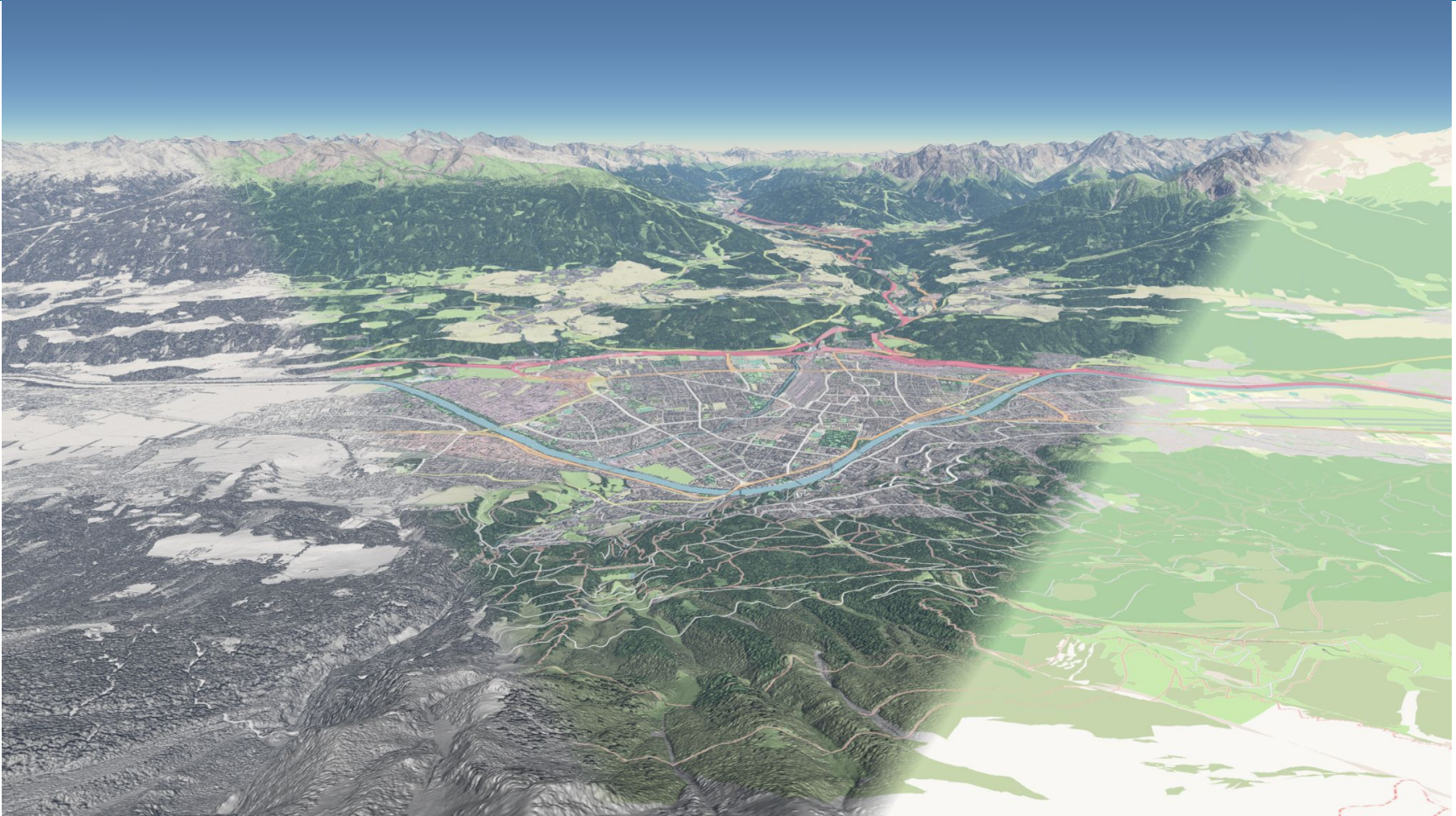


Tiles in 3D





Surface Details

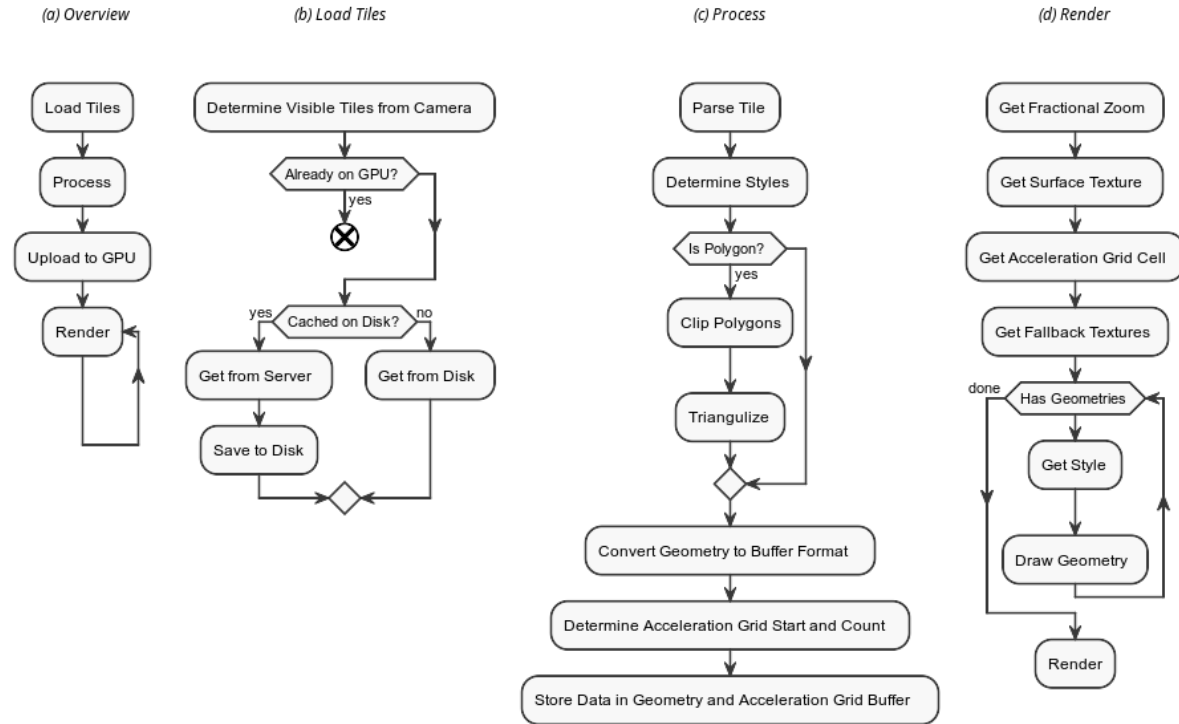


Surface Details



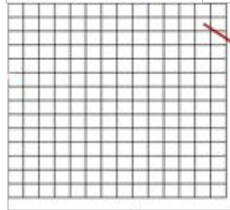


- Server - Client
- OpenMapTiles
 - Toolchain
 - OpenStreetMap
 - PostgreSQL
 - Imposm3
 - MapBox Vector Tile
- AlpineMaps.org
 - Qt Application
 - C++
 - OpenGL
 - WebAssembly



- Mapbox Vector Tile Parser
- Clipper2 - Angus Johnson
- Mapbox Earcut Triangulation Library

Acceleration Grid



Acceleration Grid Cell

24 bits	8 bits
Geometry Buffer Index	Number of Geometries

Geometry Buffer

64 bits	64 bits	64 bits	64 bits	64 bits	...
Geom 1	Geom 2	Geom 3	Geom 4	Geom 5	...

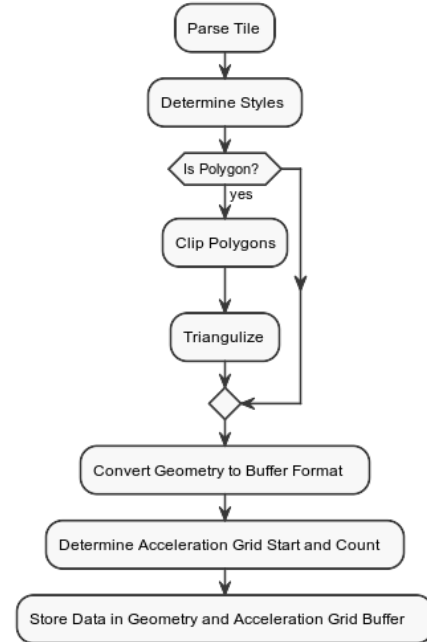
2 Geometries

Geometry Buffer Cell Polygon (2x32 bits)

8 bits	8 bits	8 bits		8 bits
x0	y0	x1		y1
8 bits	8 bits	3 bits	1 bit	12 bits
x2	y2	inner edge	is polygon	style index

Geometry Buffer Cell Line (2x32 bits)

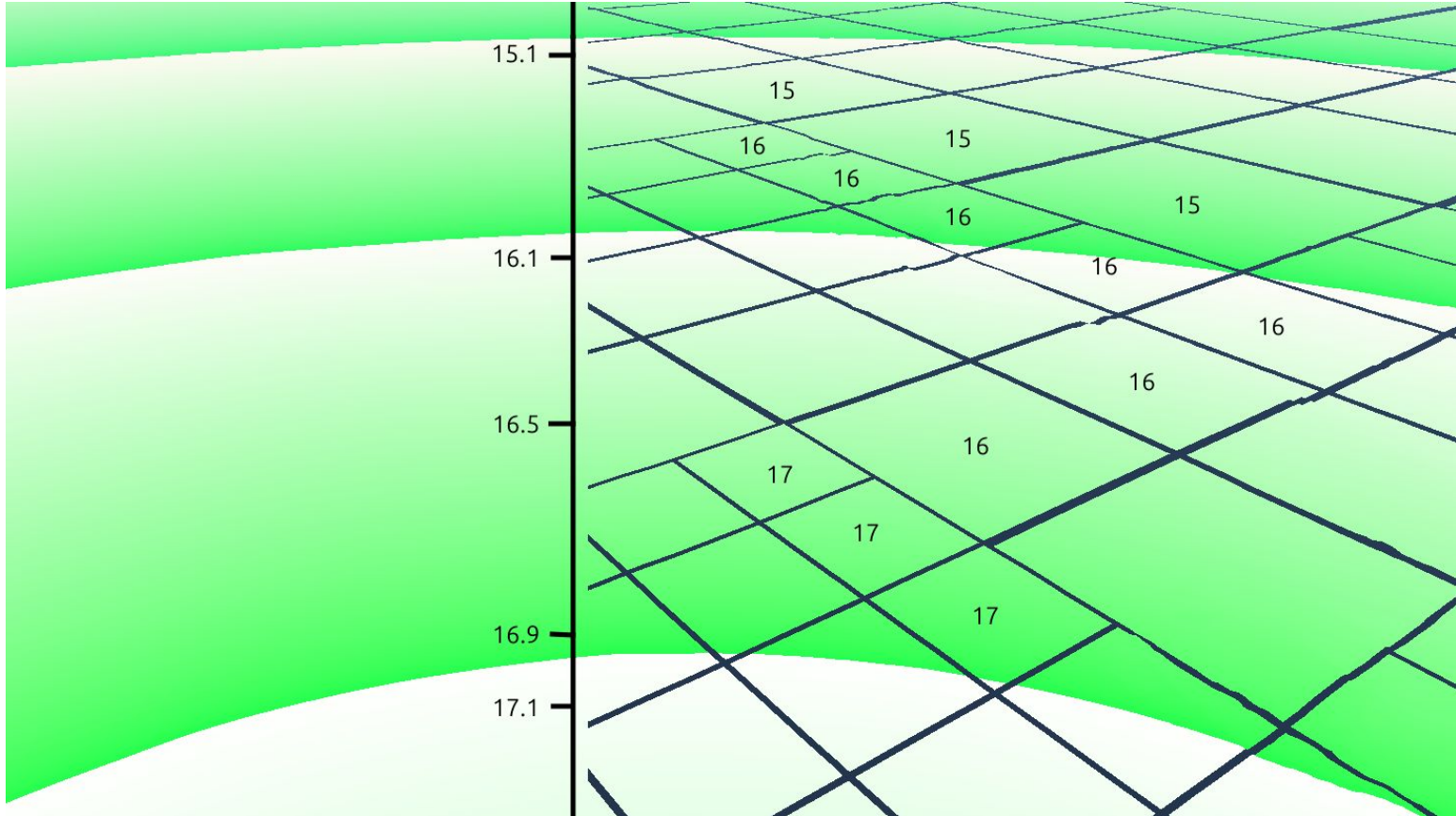
8 bits		8 bits		8 bits			8 bits
x0_0		y0_0		x1_0			y1_0
4 bits	4 bits	4 bits	4 bits	1 bit	2 bits	1 bit	12 bits
x0_1	y0_1	x1_1	y1_1	unused	line caps	is polygon	style index



Without Fractional Zoom



Fractional Zoom



- Nehab and Hoppe
 - Random-access rendering of general vector graphics [NH08]

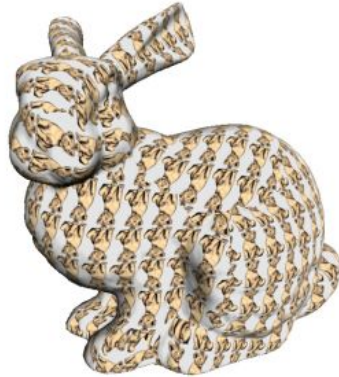
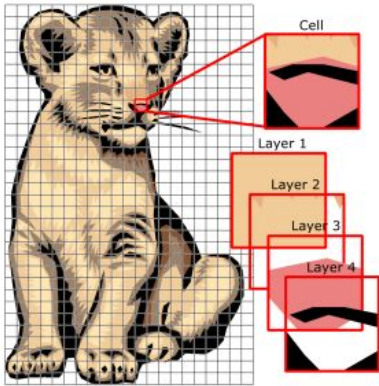
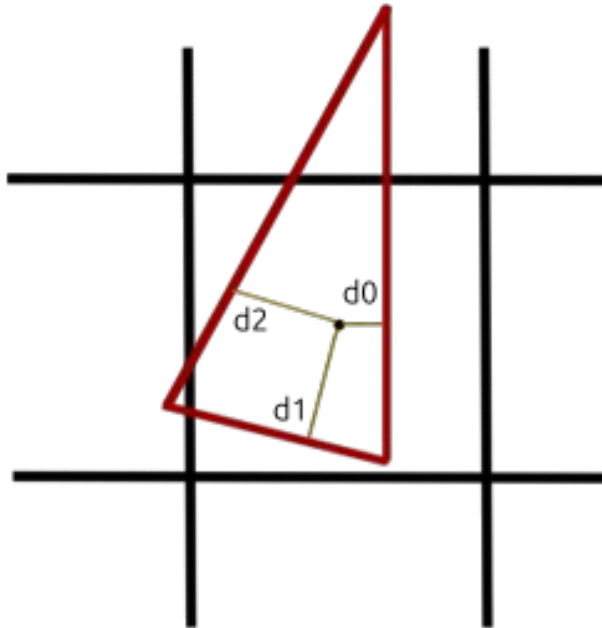


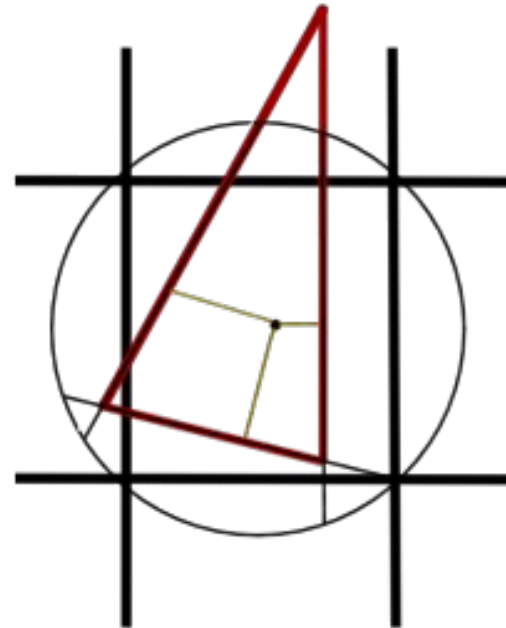
Image source: Random-access rendering of general vector graphics [NH08]



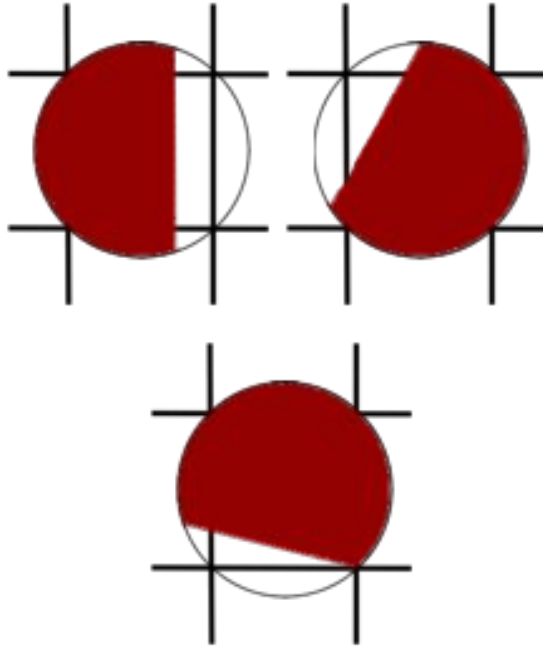
(1)



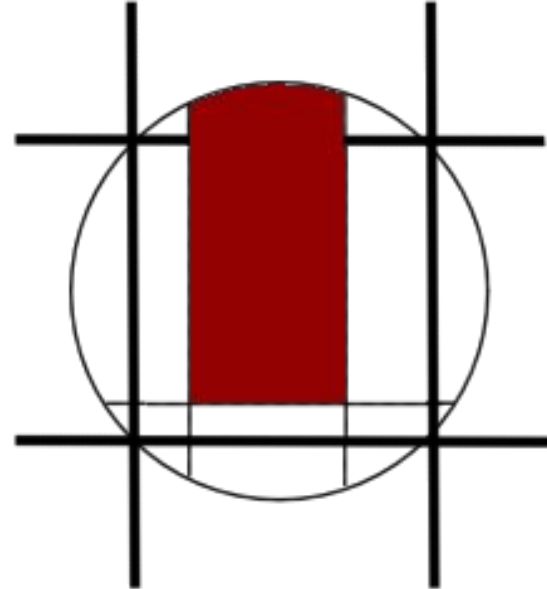
(2)



(3)



(4)



3 Half space
Complex



1 Half space
Simple reduced



1 Half space
Simple naive



Fallback Texture



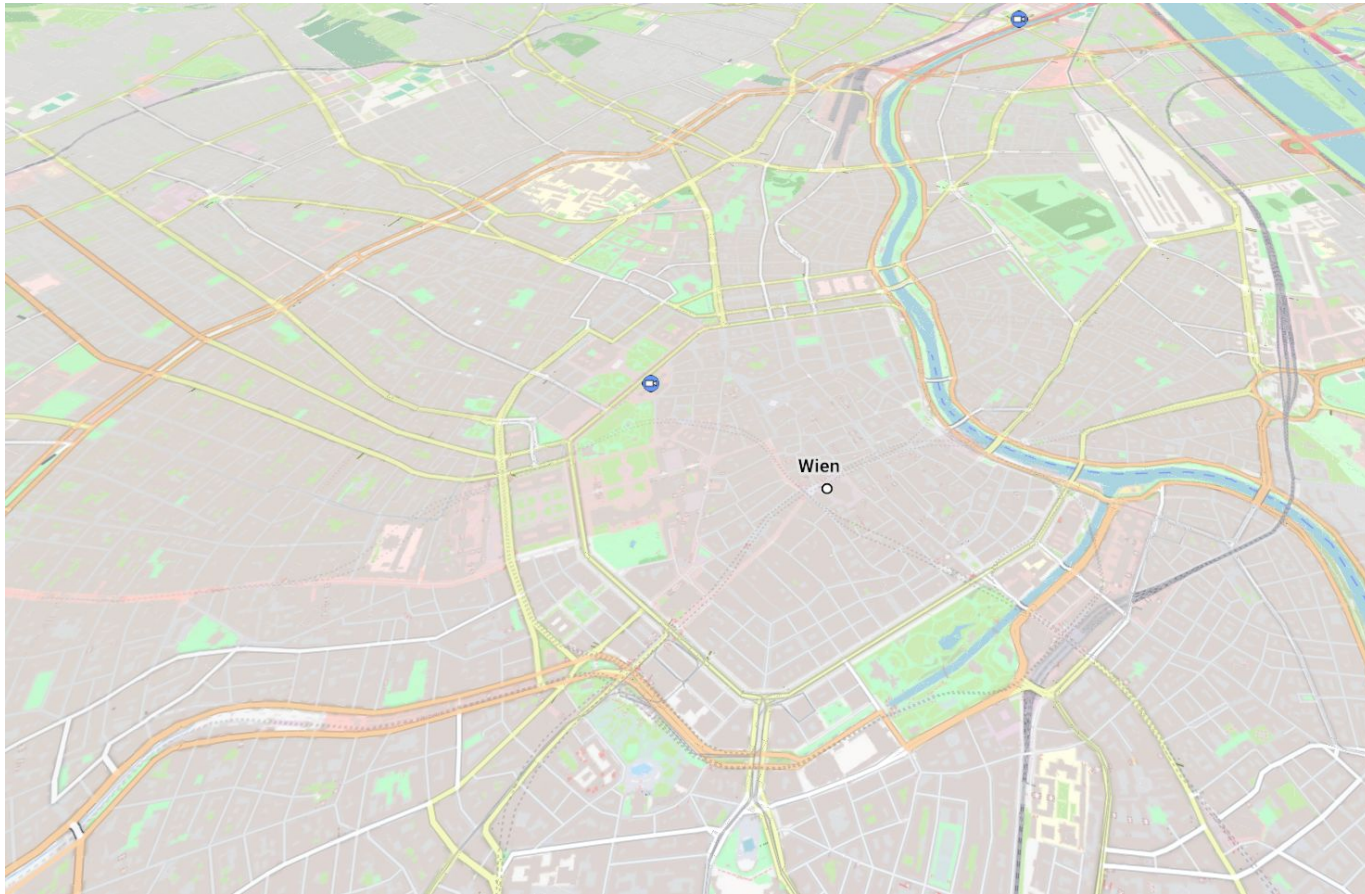
Vector Render



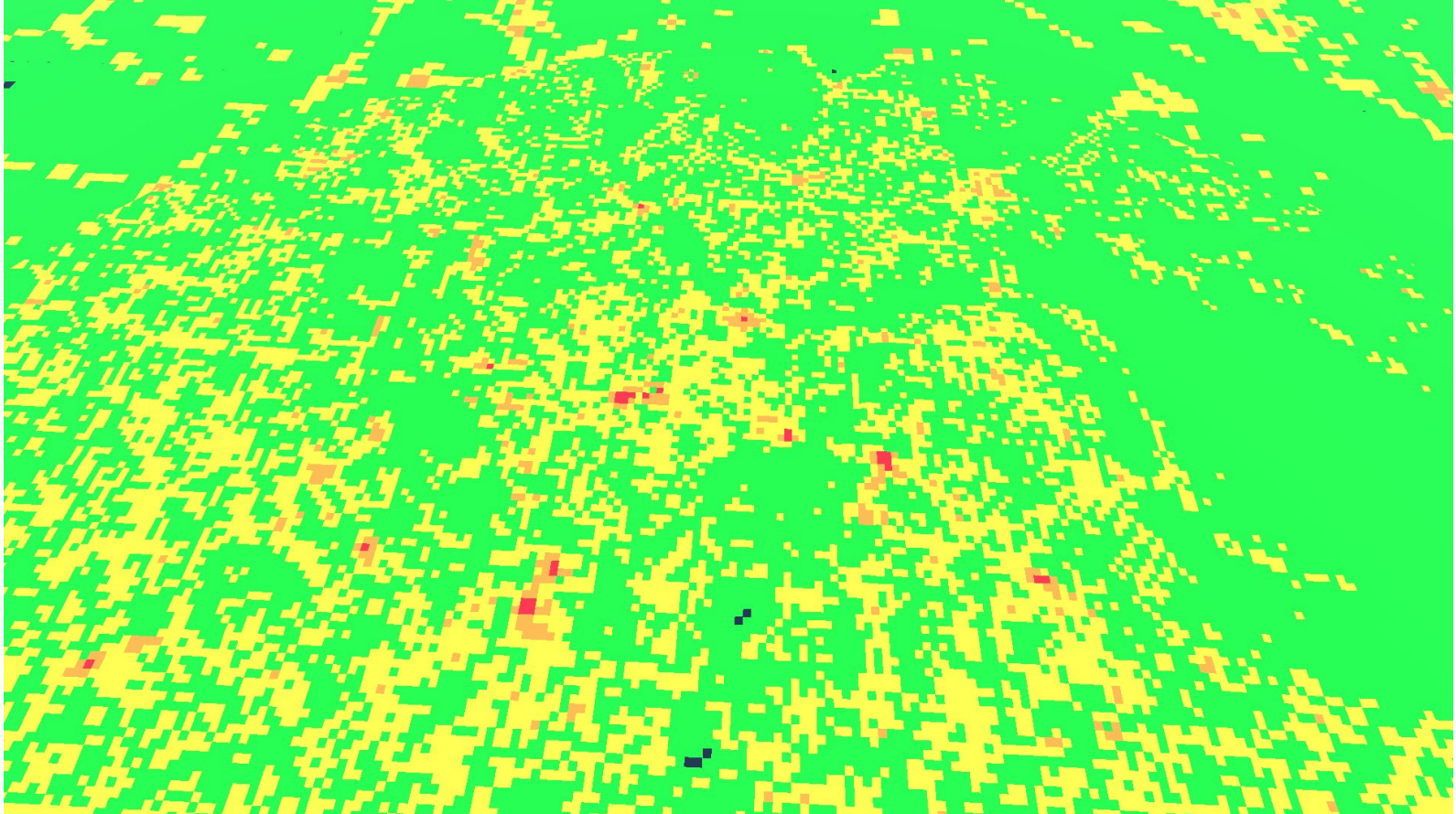
8 Geometries per Cell



Hybrid Approach



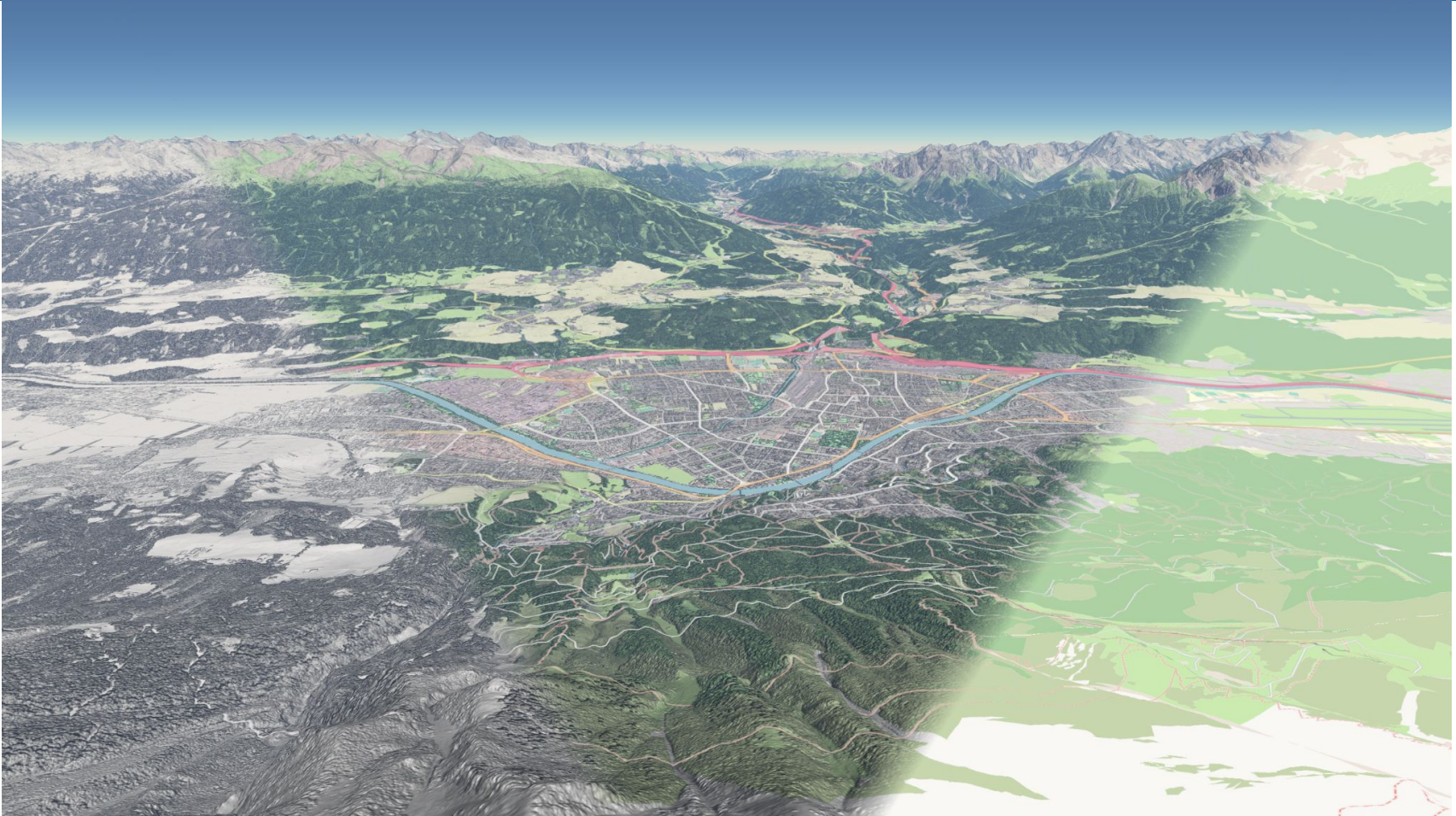
Cell Density



	Schneeberg	Innsbruck	Vienna Side	Vienna Top
Geometries	2,263,901	3,396,163	4,482,712	7,082,781
0 simple	3.05 ms	3.37 ms	3.05 ms	2.55 ms
0 complex	2.99 ms	3.34 ms	3.02 ms	2.54 ms
4 simple	5.64 ms	6.91 ms	5.94 ms	5.74 ms
4 complex	6.4 ms	7.84 ms	6.64 ms	6.52 ms
8 simple	7.54 ms	9.33 ms	8.46 ms	8.6 ms
8 complex	8.8 ms	10.87 ms	9.73 ms	10 ms
16 simple	10.02 ms	12.57 ms	12.24 ms	13.35 ms
16 complex	11.95 ms	14.99 ms	14.4 ms	15.9 ms
32 simple	11.65 ms	15.36 ms	16.3 ms	19.53 ms
32 complex	13.86 ms	18.48 ms	19.42 ms	23.59 ms
64 simple	11.98 ms	16.28 ms	19.12 ms	23.58 ms
64 complex	14.45 ms	19.61 ms	22.88 ms	28.67 ms
128 simple	12.03 ms	16.48 ms	20 ms	24.51 ms
128 complex	14.36 ms	19.84 ms	23.98 ms	29.76 ms
255 simple	12.03 ms	16.51 ms	20.04 ms	24.53 ms
255 complex	14.37 ms	19.88 ms	24.03 ms	29.81 ms



Thank You for your Attention!



- [ADP20] Alireza Amiraghdam, Alexandra Diehl, and Renato Pajarola. Localis: Locally-adaptive line simplification for gpu-based geographic vector data visualization. In Computer Graphics Forum, volume 39, pages 443–453. Wiley Online Library, 2020.
- [ADP22] Alireza Amiraghdam, Alexandra Diehl, and Renato Pajarola. Loops: Locally optimized polygon simplification. In Computer Graphics Forum, volume 41, pages 355–365. Wiley Online Library, 2022.
- [Cur22] Chris Currier. Protocol buffers. In Mobile Forensics—The File Format Handbook: Common File Formats and File Systems Used in Mobile Devices, pages 223–260. Springer, 2022.
- [Gam95] Erich Gamma. Design patterns elements of reusable object-oriented software, chapter 4 Structural Patterns, pages 183–195. Addison-Wesley, 1995.
- [git26a] GitHub - mapbox/earcut.hpp: Fast, header-only polygon triangulation — github.com. <https://github.com/mapbox/earcut.hpp>, 2026. [Accessed 05-07-2026].
- [git26b] GitHub - mapbox/vector-tile: C++14 library for decoding Mapbox Vector Tiles — github.com. <https://github.com/mapbox/vector-tile>, 2026. [Accessed 05-07-2026].
- [imp] imposm.org - openstreetmap to postgis importer. <https://imposm.org/>. [Accessed 09-12-2025].
- [Joh26] Angus Johnson. Clipper2 - Polygon Clipping Offsetting & Triangulating — angusj.com. <https://www.angusj.com/clipper2/{D}ocs/{O}verview.htm>, 2026. [Accessed 05-07-2026].
- [mapa] MapLibre Style Spec — maplibre.org. <https://maplibre.org/maplibre-style-spec/>. [Accessed 10-12-2025].



- [mapb] Vector tiles standards | Tilesets | Mapbox Docs — docs.mapbox.com. <https://docs.mapbox.com/data/tilesets/guides/vector-tiles-standards/>. [Accessed 09-12-2025].
- [NH08] Diego Nehab and Hugues Hoppe. Random-access rendering of general vector graphics. *ACM Transactions on Graphics (TOG)*, 27(5):1–10, 2008.
- [ope] OpenStreetMap — openstreetmap.org. <https://www.openstreetmap.org>. [Accessed 05-12-2025].
- [QMK08] Zheng Qin, Michael D McCool, and Craig Kaplan. Precise vector textures for real-time 3d rendering. In *Proceedings of the 2008 symposium on Interactive 3D graphics and games*, pages 199–206, 2008.
- [Ste17] Emmanuel Stefanakis. Web mercator and raster tile maps: two cornerstones of online map service providers. *Geomatica*, 71(2):100–109, 2017.
- [TBP16] Matthias Thöny, Markus Billeter, and Renato Pajarola. Deferred vector map visualization. In *SIGGRAPH ASIA 2016 symposium on visualization*, pages 1–8, 2016.
- [TBP18] Matthias Thöny, Markus Billeter, and Renato Pajarola. Large-scale pixel-precise deferred vector maps. In *Computer Graphics Forum*, volume 37, pages 338–349. Wiley Online Library, 2018.
- [vmv16] Deferred vector map visualization, 2016.
- [Zec20] Silvana Zechmeister. Interaktive Visualisierung von Vektordaten auf Höhenfeldern. PhD thesis, Wien, 2020.





- MapBox Vector Tile standard (version 2.1)
- Google's Protocol Buffers
- Type
 - Polygon, Polyline, Point
- Layer
 - Landuse, Landcover, Transportation, ...
- Attributes
 - Class, Subclass, Ramp, Bicycle, Brunnel, ...
- Geometry
- Zoom



- MapLibre Style Specification
- Attributes
 - Line Cap
 - Line Ending
 - Color
 - Line Width
 - Opacity
- Zoom Interpolation
- 1 Geometry -> n Styles

```
"layers":[
{
  "id": "landcover_subclass_outline",
  "type": "line",
  "source-layer": "landcover",
  "minzoom": 11,
  "maxzoom": 18,
  "layout":{
    "line-cap": "round"
  },
  "paint":{
    "line-color":[
      "match",
      ["get", "subclass"],
      "wood", "#add19e",
      "rock", "#eee5dc",
      "#000"
    ],
    "line-width":{
      "base": 1.2,
      "stops":
      [
        [11, 1], [13, 4], [18, 11]
      ]
    }
  },
  "filter":[
    "all",
    [
      "in",
      "subclass",
      "wood",
      "rock"
    ]
  ]
},
]
```

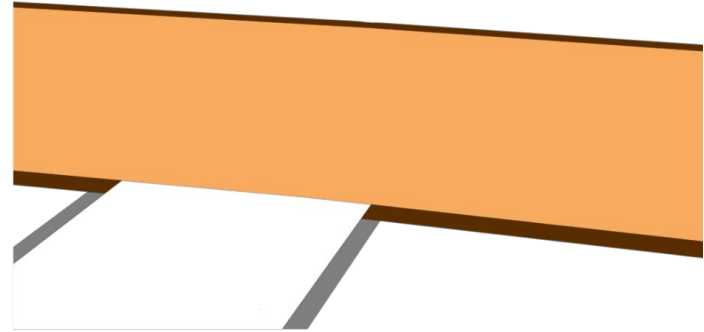
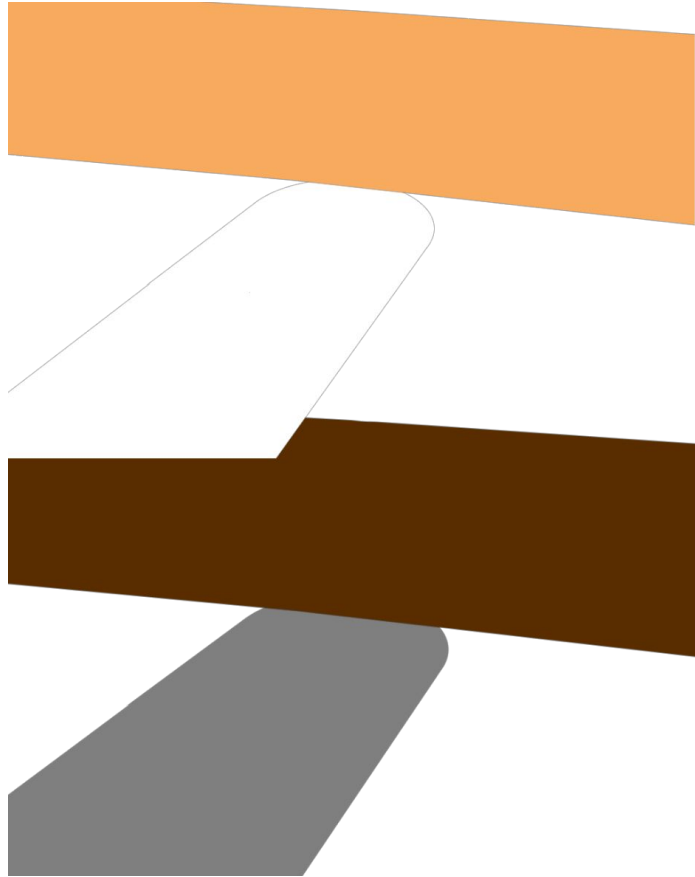


Tile Geometry	Style
Zoom	Zoom
Type	Type
Layer	Source-Layer
Attributes	Filter
1 Geometry	n Styles

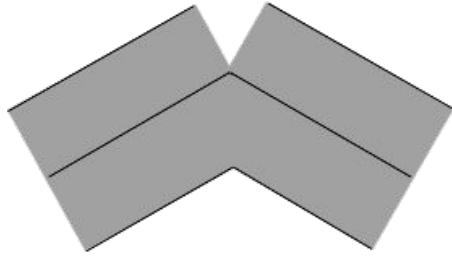
```
"filter": [
  "all",
  [
    "any",
    [
      [
        "==",
        "class",
        "minor"
      ],
      [
        "==",
        "class",
        "major"
      ]
    ],
    [
      "!=",
      "ramp",
      true
    ]
  ]
]
```



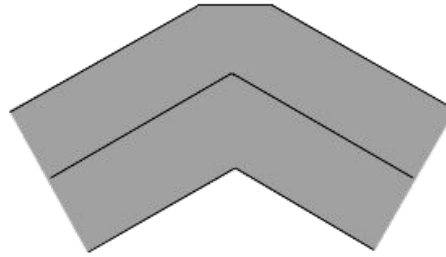
Layers



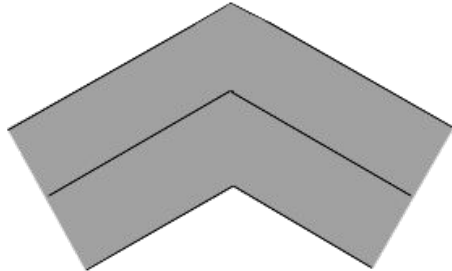
without line joins



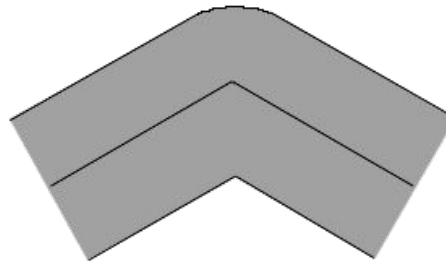
bevel



miter



round



Butt



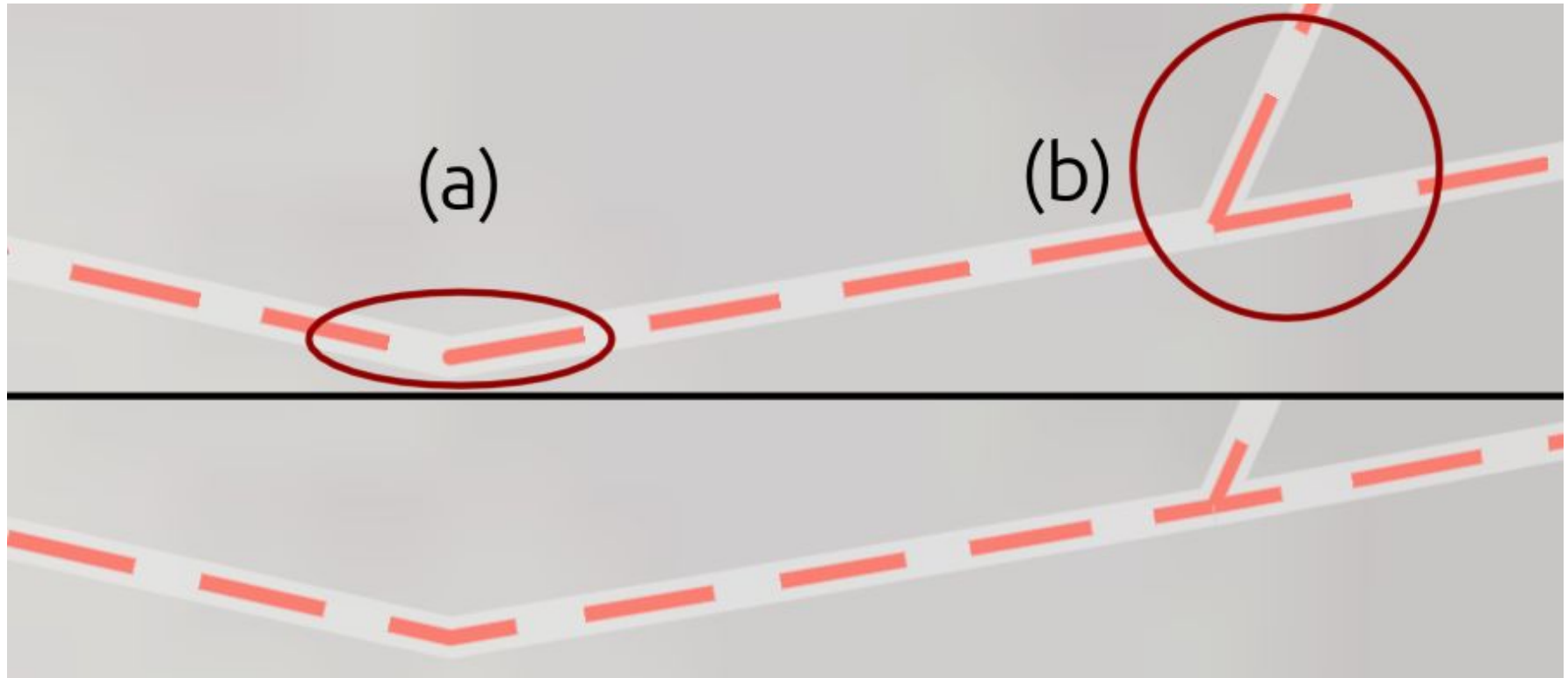
Square



Round



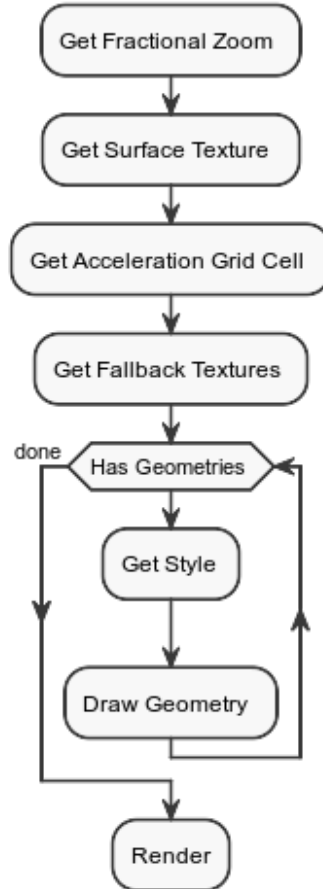
Line Endings and Line Joins



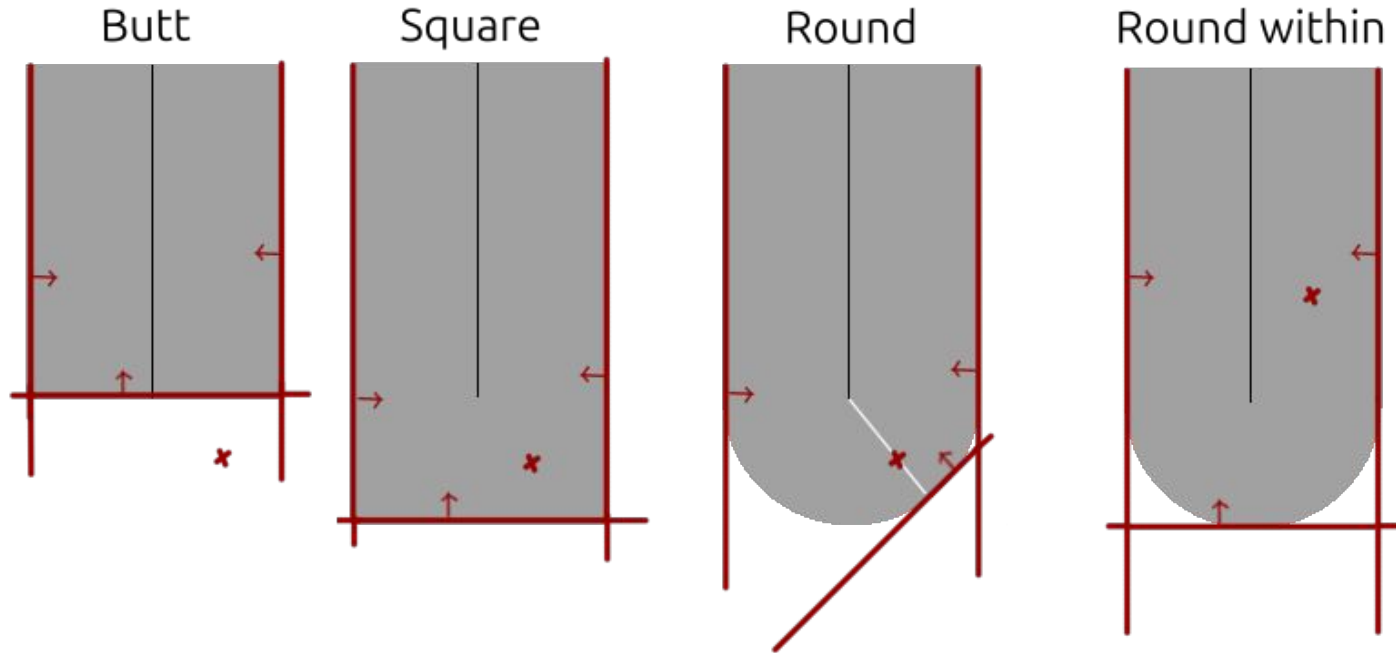
- Instanced Drawing
 - 1 Instance = 1 Tile
- Calculates:
 - UV
 - Worldspace position
 - Normal
 - Tile id



Fragment Shader



Line Coverage



Buffer Overview

Indirection Map

Zoom Offset	Instance 1	Instance 2	Instance 3	Instance 4	Instance 5	...
0	Layer 113, 216, 0	Layer 48, 88, 1	Layer 102, 104, 0	Layer 214, 314, 0	Layer 399, 199, 0	...
1	Layer 368, 165, 1	Layer 388, 988, 0	Layer 171, 175, 2	Layer 397, 393, 2	Layer 542, 342, 2	...
2	Layer 423, 222, 2	Layer 21, 20, 2	Layer 150, 152, 1	Layer 435, 433, 0	Layer 410, 416, 2	...
3	Layer 139, 636, 1	Layer 254, 252, 0	Layer 158, 452, 0	Layer 405, 205, 1	Layer 729, 329, 1	...

Acceleration Grid

Layer 158	0, 90	90, 38	128, 2	...
	480, 22	502, 10	512, 2	...
	...	...	...	...
	...	...	...	...
	...	...	...	...
Layer 159	0, 55	55, 20	...	...
	...	...	...	...
	...	...	...	...
	...	...	...	...
Layer ...	0, 20	20, 45	...	...
	...	...	...	...
	...	...	...	...

Geometry Buffer 0 (128x128)

Layer 452	Geometry 1	Geometry 2	...
	Geometry 128	Geometry 129	...
	...	...	...
	...	...	...
	...	...	...
Layer ...	Geometry 1	Geometry 2	...
	...	...	...
	...	...	...

Acceleration Grid Entry

Offset in Geometry Buffer	Number of Geometries
---------------------------	----------------------

Geometry Entry

Vertices	Metadata	StyleIndex
----------	----------	------------

Style Buffer

Style 1
Style 2
Style ...

Surface Map Buffer

Layer 158
Layer 159
Layer ...

Terrain Height Buffer

Layer 102
Layer 103
Layer ...

Geometry Buffer 1 (256x256)

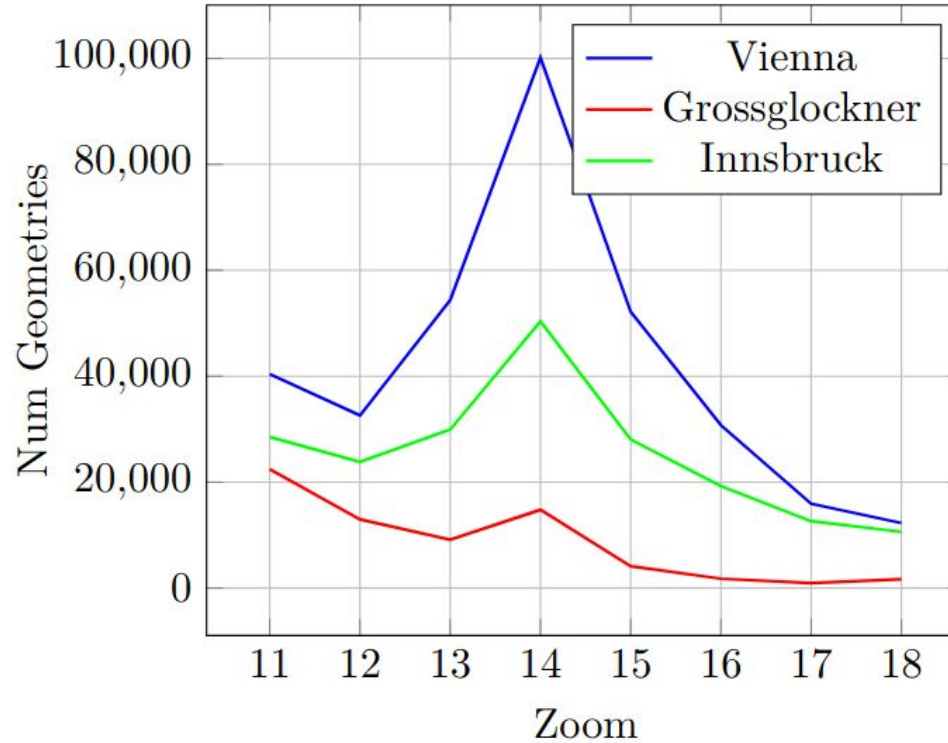
Layer 452	Geometry 1	Geometry 2	...
	Geometry 256	Geometry 257	...
	...	...	...
	...	...	...
	...	...	...

Geometry Buffer 2 (512x512)

Layer ...	Geometry 1	Geometry 2	...
	...	...	...
	...	...	...



Geometries per Zoom



Visible Tiles:

$$P_e \leq \frac{V_y}{2 \tan c_{fov}} \frac{\sqrt{2} T_w}{T_d T_p} \quad (3.1)$$

Equation 3.1 symbol definition: P_e is a pixel error threshold: a value of 0.5 means that the tile is accurate up to half a pixel, V_y is the vertical viewport size, c_{fov} is the camera field of view in radians, T_w is the tile size in world space, T_d is the distance to the nearest part of the tile and T_p is the resolution of the tile in pixels.

Fractional Zoom:

$$z = \log_2 \left(\frac{V_y T_w \sqrt{2}}{2 \tan(c_{fov}) C_d P_e} \right) \quad (3.2)$$

Equation 3.2 symbol definition: z is the fractional zoom level, V_y is the vertical viewport size, C_d is the distance between camera and evaluation point, T_w is the tile size in world space and P_e is a pixel error threshold: a value of 0.5 means that the tile is accurate up to half a pixel,



$$frag_{rgba} = frag_{rgba} + (1 - frag_a) * layer_a * layer_{rgba}$$



Perspective Line Width

